

HIGH WYCOMBE, United Kingdom

WMO: 036600

Lat: 51.6817N

Lon: 0.8069W

Elev: 673

StdP: 14.34

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	28.5	21.0	16.1	31.6	23.4	18.1	31.6	21.0	46.5	19.3	46.6	6.9	300	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	79.1	64.8	75.3	63.1	72.1	61.7	66.4	75.8	64.6	72.5	63.0	69.6	6.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	88.3	69.1	61.6	83.9	67.3	60.2	79.9	65.7	31.3	75.5	29.9	72.6	28.8	69.7	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.9	14.2	24.0	84.8	2.8	4.7	22.0	88.2	20.4	91.0	18.8	93.7	16.8	97.1		
(5)			23.1	70.2	2.8	2.9	21.1	72.4	19.4	74.1	17.8	75.7	15.8	77.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	39.7	40.2	43.4	47.7	53.2	58.3	62.1	61.4	57.6	51.5	44.6	40.9		
	DBStd	9.62	5.55	5.76	5.63	5.65	5.31	4.79	4.92	4.39	4.43	4.79	5.27	6.09		
	HDD50	1443	320	277	212	107	25	1	0	0	2	38	174	287		
	HDD65	5504	784	695	669	520	368	209	120	130	226	421	613	749		
	CDD50	1478	1	2	8	37	123	250	376	354	230	83	11	3		
	CDD65	63	0	0	0	0	1	8	31	19	4	0	0	0		
	CDH74	455	0	0	0	2	9	71	231	120	20	2	0	0		
	CDH80	76	0	0	0	0	0	10	43	23	0	0	0	0		
(14) Wind	WSAvg	7.2	8.3	8.0	8.0	7.4	7.1	6.5	6.2	6.1	6.4	7.5	7.3	8.2		
(15) Precipitation	PrecAvg	26.5	2.6	1.9	1.6	2.0	2.0	2.0	1.9	2.4	2.2	2.6	2.9	2.4		
	PrecMax	34.6	6.1	4.9	3.9	5.4	5.3	5.4	4.4	5.6	4.5	5.7	5.7	4.4		
	PrecMin	19.5	0.2	0.0	0.5	0.0	0.2	0.4	0.6	0.3	0.5	0.8	1.2	0.5		
	PrecStd	4.6	1.3	1.2	0.9	1.3	1.1	1.2	1.0	1.3	1.2	1.3	1.3	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	56.3	61.8	71.0	75.0	81.1	84.6	82.8	76.9	67.0	58.4	55.3		
		MCWB	51.6	50.2	51.0	56.7	62.0	65.7	67.0	67.7	63.7	60.3	56.3	52.9		
	2%	DB	51.9	52.2	57.0	64.6	70.5	75.2	80.0	77.0	71.8	62.3	55.9	53.2		
		MCWB	49.9	48.6	49.5	54.3	59.9	63.0	65.2	64.9	61.3	57.9	53.7	51.5		
	5%	DB	49.9	50.1	53.9	60.2	66.7	71.2	76.0	72.9	67.9	60.1	54.1	51.7		
		MCWB	48.0	47.2	48.5	51.8	57.8	61.3	63.6	62.6	59.8	56.6	52.2	50.0		
	10%	DB	48.0	48.2	51.4	56.6	62.6	67.7	72.1	69.5	65.0	58.3	52.3	49.8		
		MCWB	46.4	45.9	47.3	49.8	55.2	59.3	62.0	60.9	58.5	55.6	50.4	48.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	51.8	53.1	58.8	64.4	67.7	69.1	69.3	66.1	61.6	57.0	53.7		
		MCDB	53.1	54.5	57.5	68.6	71.5	78.7	80.9	80.7	72.3	65.4	57.9	54.4		
	2%	WB	50.2	49.5	51.1	55.3	61.0	64.4	66.4	65.8	63.3	59.1	54.4	52.1		
		MCDB	51.5	51.3	54.8	63.4	68.2	72.2	76.8	73.9	69.1	61.1	55.3	53.1		
	5%	WB	48.3	47.8	49.7	52.9	58.5	62.1	64.6	63.8	61.5	57.6	52.5	50.3		
		MCDB	49.5	49.5	52.6	58.4	65.0	68.9	73.6	70.3	65.9	59.2	53.8	51.4		
	10%	WB	46.6	46.1	48.1	50.9	56.4	60.3	62.9	62.1	59.8	56.2	50.7	48.2		
		MCDB	47.9	47.9	51.1	55.6	61.6	66.7	70.4	67.4	63.6	57.9	52.0	49.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	11.1	13.7	14.3	14.9	15.5	14.6	13.3	10.3	8.4	7.6		
		MCDBR	9.8	10.4	14.2	18.6	19.5	20.6	21.6	19.5	16.9	11.4	9.4	8.6		
		MCWBR	9.8	8.0	8.9	10.7	11.1	11.0	10.0	9.4	8.6	7.7	8.3	8.7		
	5% WB	MCDBR	9.6	10.0	12.6	16.8	17.7	18.3	19.0	16.8	15.0	10.5	9.3	8.7		
		MCWBR	9.8	8.7	8.9	10.3	10.7	10.6	9.9	9.1	9.1	8.3	8.9	9.0		
(40) Clear-Sky Solar Irradiance	taub	0.322	0.338	0.372	0.394	0.398	0.397	0.393	0.383	0.366	0.350	0.324	0.310			
	taud	2.373	2.386	2.282	2.237	2.262	2.289	2.307	2.354	2.402	2.430	2.418	2.356			
	Ebn at Noon	217	248	259	267	271	272	271	268	259	239	214	199			
	Edh at Noon	20	25	34	40	40	40	38	35	30	24	19	17			
(44) All-Sky Solar Radiation	RadAvg	256	465	806	1240	1509	1659	1576	1308	986	574	305	202			
	RadStd	21	60	108	144	96	169	160	105	85	51	35	24			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page